

Interpretable EEG Eye-State Detection: Comparing Feature Representations under Temporally Rigorous Cross-Validation

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Abstract

Electroencephalography (EEG) measures electrical activity of the brain through scalp electrodes, enabling non-invasive monitoring of neural states. EEG eye-state classification is the task of detecting whether a subject's eyes are open or closed from scalp electrode recordings. This paper addresses limitations of current EEG datasets, such as temporal data leakage, SNR constraints of consumer-grade hardware, and the resulting limitations on spectral feature extraction. It used the UCI EEG Eye State dataset, which was recorded with an Emotiv Epoc headset. I preprocessed the data in sliding windows, performed per-window mean subtraction, artifact rejection, blocked cross-validation, and performed feature extraction of spectral, statistical, and raw EEG data on Random Forest, LSTM, and transformer models. The LSTM outperformed all Random Forest models with 0.68 ROC-AUC. A subsequent bidirectional Transformer experiment achieved 0.70 ROC-AUC using RoPE positional encoding and window-level instance normalization. Furthermore, diagnostic attention heatmaps provided interpretable visual proof of token-to-token dependencies, demonstrating that temporal modeling with unmasked attention captures discriminative structure that handcrafted features miss, provided methodological rigor in train-test splitting is maintained.

1. Introduction

EEG-based eye-state classification has practical applications in drowsiness detection for driver safety monitoring and in the development of brain-computer interfaces that enable hands-free control for users with motor impairments.

The alpha rhythm is the relaxed state of eye closure, characterized by 8-13 Hz oscillations originating in occipital regions, and it disappears when eyes open because the visual cortex shifts from a resting to active state. This makes EEG eye-state classification theoretically tractable because the alpha rhythm provides a reliable neural correlate of eye state that should be detectable from scalp electrodes.

Most existing work uses random train/test splits that inflate performance through data leakage. Consumer-grade hardware introduces noise that challenges spectral methods (Nolan et al., 2010). The UCI EEG Eye State dataset, recorded using a consumer-grade Emotiv Epoc headset, provides a standard benchmark for evaluating these challenges.

This paper presents an evaluation of multiple feature representations and classification architectures for EEG eye-state detection using the UCI EEG Eye State dataset. Random Forest models of different features were compared to an LSTM on raw EEG, with a Transformer experiment extending the comparison. Results demonstrated that the LSTM outperformed all Random Forest models, and the subsequent optimized Transformer further improved generalization, achieving 0.70 ROC-AUC. Individual temporal blocks also peaked at 0.84+ ROC-AUC, but the variance is an honest reflection of drift during the 2-minute session. These findings suggest that attention over raw temporal windows captures discriminative structure that handcrafted features may miss, provided the model capacity is strictly regularized and domain shifts are properly managed.

2. Methods

2.1 Dataset and Preprocessing

The dataset was a 117-second continuous recording of EEG eye states. There were 14980 samples (meaning around 128 samples per second) and 14 channels. Each label is either a "0," indicating eyes open, or a "1," indicating eyes closed. Data was preprocessed in sliding windows of size 128 and stride 32. Per-window mean subtraction was implemented to remove DC drift. Artifact rejection was then implemented; windows where any channel exceeded 150 μ V peak amplitude were discarded.

For the Transformer, initial global fold z-scoring across the dataset caused severe waveform distortion across time. Instead, window-level instance normalization (standardizing every 128-timestep window independently to zero-mean and unit-variance per channel). This eliminated baseline drift, reduced cross-fold variance from 0.137 to 0.115, and stabilized model generalization.

Train and test sets were split through 5-fold Stratified Group K-Fold (SGKF) cross-validation rather than a random split or standard blocked cross-validation. To prevent the model from training on data that is temporally adjacent to test data (overlap leakage), the continuous recording was divided into contiguous macro-blocks of 1000 samples. SGKF kept windows from the same macro-block together while shuffling the blocks to ensure each fold maintained a representative balance of eyes-open and eyes-closed states. This resolved the severe class imbalance issues inherent to standard sequential splitting on non-stationary EEG recordings. StandardScaler was fitted on training data only per fold.

2.2 Feature Extraction

A real-valued Fast Fourier Transform (rfft) extracted power spectrum bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), and beta (13-31 Hz). Power was averaged per channel, log-compressed, yielding 56 features. Statistical features (mean, std, min, max, skewness, kurtosis) computed per channel added 84 features. Additionally, an alpha/beta ratio feature was

calculated (14 features). Feature representations were evaluated independently and concatenated (140 features) to assess spectral vs. time-domain contributions.

2.3 Models

A Random Forest model of 100 estimators and random state 42 was used. It received inputs of flattened feature vectors. There was no temporal information used, as each window was treated independently. An LSTM with a single layer, hidden size of 32, and dropout of 0.3 was trained on raw EEG windows of shape (batch, 128, 14) using BCEWithLogitsLoss, the Adam optimizer with learning rate 1e-3, and gradient clipping at max_norm = 1.0. It trained for 30 epochs, determined empirically to minimize overfitting. The LSTM is architecturally suited for EEG because it processes sequential data and maintains temporal context across timesteps, unlike Random Forest which treats each window as an orderless feature vector.

To establish a rigorous baseline for attention mechanisms, a 2-layer transformer with 16-dimensional embeddings and 4 attention heads was trained on raw EEG windows of shape (batch, 128, 14). Initial experiments utilized an autoregressive casual mask (standard in language models), which produced "attention sinks" on timestep 0 and failed to capture the structure of a whole window. Removing lower-triangular masking converted the architecture into a Bidirectional Encoder, allowing attention to correlate temporal features across the entire 128-timestep sequence simultaneously. Hyperparameters were systematically optimized using a Bayesian search (Optuna TPE with Median Pruning). The model was trained using an AdamW optimizer with weight decay 0.021, dropout of 0.358, and a learning rate of 1.78e-4. It utilized RoPE positional encoding, label smoothing (0.1), and patience-based early stopping monitored via an Exponential Moving Average of validation loss.

3. Results

3.1 Model Comparison

Model	Features	ROC-AUC	Std
Random Forest	FFT Bandpower	0.52	0.065
Random Forest	Alpha/Beta Ratio	0.46	0.058
Random Forest	Combined	0.59	0.039
Random Forest	Statistical	0.62	0.028
LSTM	Raw EEG	0.68	0.082
Transformer	Raw EEG (Windowed + Instance Norm)	0.70	0.115

Table 1: Model comparison by ROC-AUC

The LSTM scored 0.68 ± 0.082 ROC-AUC, outperforming all Random Forest models. The optimized Bidirectional Transformer achieved 0.70 ± 0.115 ROC-AUC under temporally rigorous evaluation, outperforming all other baselines. Peak individual subject folds on the Transformer reached up to 0.84 ROC-AUC.

3.2 Post-Hoc Feature Interpretability (SHAP)

SHAP analysis on the Random Forest revealed that max and kurtosis of frontal and temporal electrodes dominated predictions, rather than the expected occipital alpha rhythm. High kurtosis at frontal electrodes strongly predicted eyes-open, indicating that eye movements generate spike-like artifacts. This suggests the discriminative signal in this dataset derives primarily from oculomotor artifacts rather than alpha suppression, constrained by the Emotiv Epoc's poor occipital signal quality.

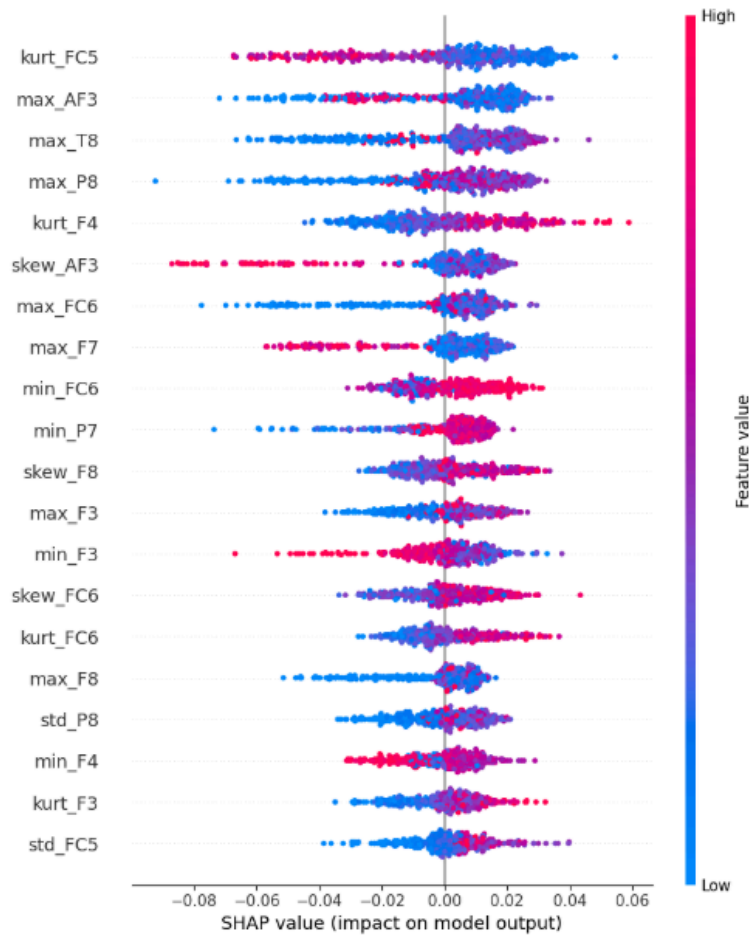


Figure 1: SHAP summary plot showing feature importance for the Random Forest statistical model. Features are ranked by mean absolute SHAP value. Pink indicates high feature values, blue indicates low feature values.

3.3 Native Attention Interpretability

Unlike SHAP, which only identifies *which* electrodes are important, extracting the attention weight matrices from the Transformer revealed *how* the network processes temporal signals. Visualization of Layer 0 Attention Heatmaps demonstrated temporal attention dynamics. Heads 0 and 2 show clear diagonal frequency banding across the entire window, meaning the model captured fixed relative phase lags across subsequent queries ($j - i = \Delta t$). All heads show vertical banding, which indicate sharp voltage gradients or artifact boundaries that were generally prominent across queries and highlights attention sinks. They also showed horizontal banding, meaning some queries had uniform distributions across the sequence instead of capturing certain ones sharply. Meanwhile, white spots, such as columns 30-50 in Head 1, signal regions of keys that were not particularly important and were suppressed by the model.

Layer 0 Attention Heatmaps (Eyes Open State)

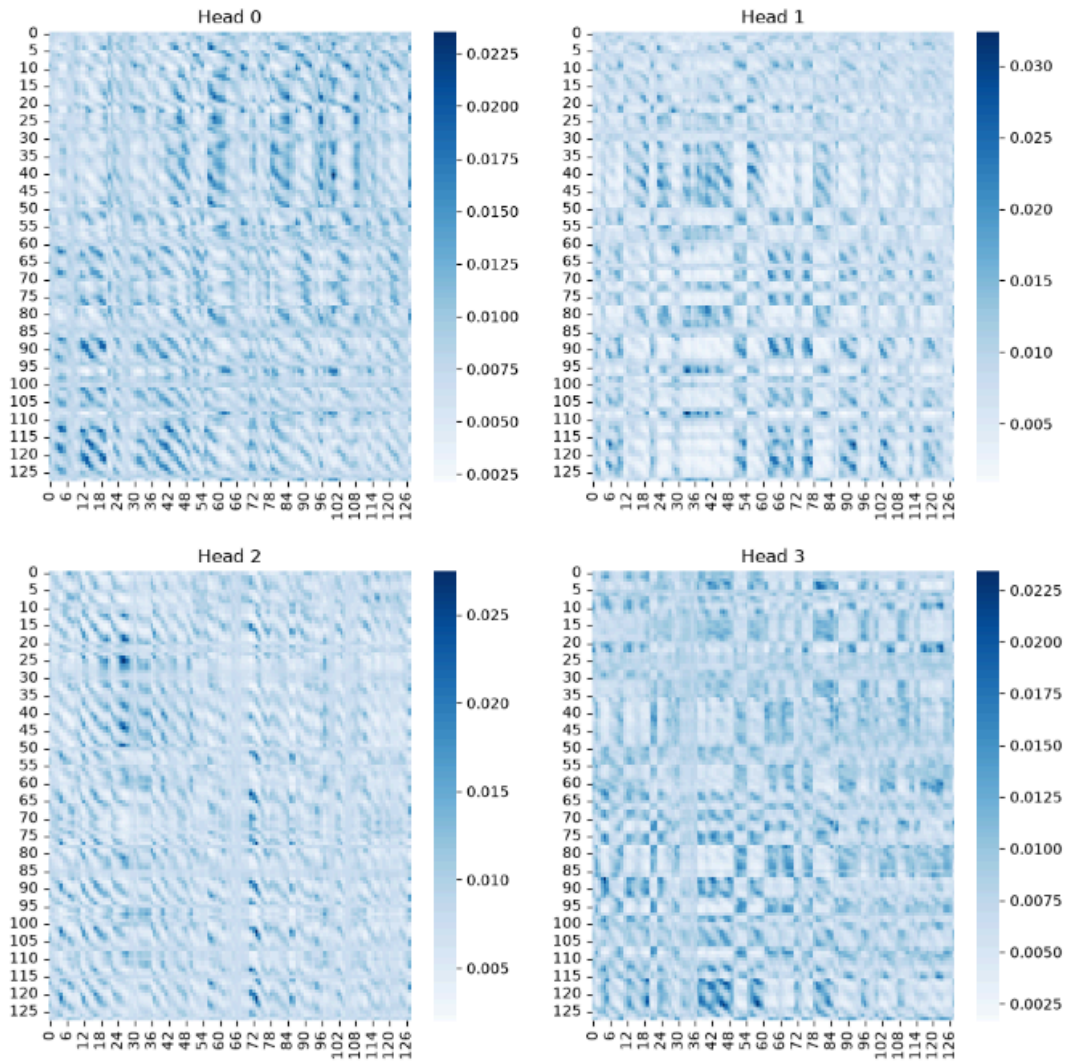


Figure 2: Layer 0 Attention Heatmaps extracted from the trained Transformer during an eyes-open state. Rows represent queries, and columns represent keys.

4. Discussion

Spectral features performed poorly because the alpha power difference was insufficient for reliable classification given the SNR limitations of consumer-grade hardware. Statistical features outperformed them by capturing global signal dynamics without depending on precise frequency resolution. The LSTM and Transformer outperformed Random Forest because they maintained temporal context across the full 128-sample window.

The transition to a Bidirectional Transformer coupled with Window-Level Instance Normalization was critical. Global z-scoring causes severe waveform distortion across time. Instance Normalization eliminated this domain shift, reducing cross-fold variance. The diagnostic attention heatmaps prove that the network actively learned to track frequencies and isolate artifacts, validating the Transformer's capacity for raw EEG classification without spatial convolutional pooling.

Future work should address the limitations of single-subject consumer-grade recordings by evaluating this bidirectional architecture on multi-subject, clinical-grade EEG datasets to fully leverage its temporal modeling capacity.

5. Conclusion

This paper compared and evaluated multiple classification models for EEG eye-state detection using the UCI EEG Eye State dataset. The LSTM outperformed all Random Forest models, while statistical features were the best-performing Random Forest. This meant the SNR limitations of consumer-grade hardware and single-subject recording constrained overall classification performance, indicating future work to aim towards multi-subject recordings and clinical-grade EEG hardware to provide sufficient data. Subsequently, an optimized transformer extension implementing RoPE, window-level instance normalization, and Stratified Group K-Fold cross-validation achieved 0.70 ROC-AUC, outperforming the LSTM baseline. Diagnostic heatmaps confirmed the model successfully learned interpretable frequency-tracking and artifact-broadcasting behaviors natively within its attention heads, suggesting bidirectional Transformers warrant extensive investigation on clinical-grade BCI datasets.

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